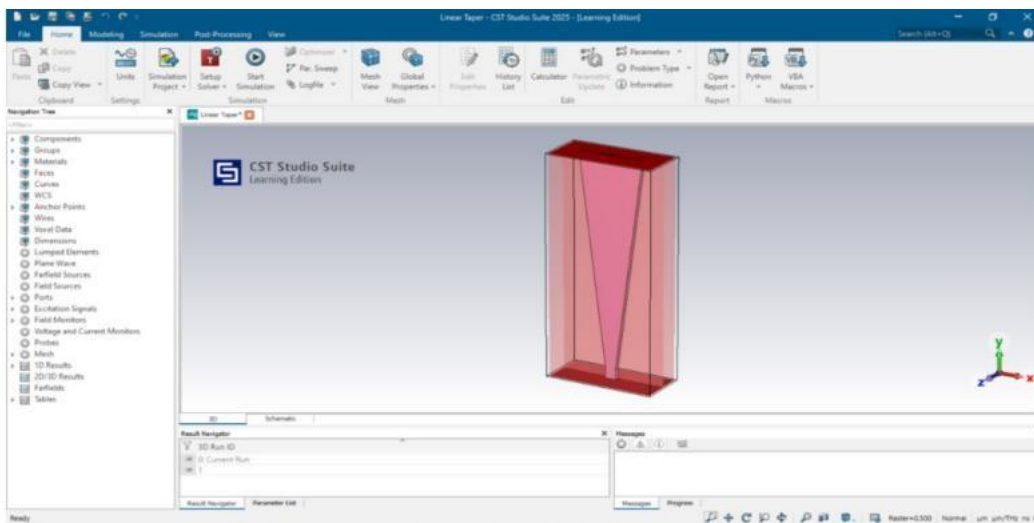


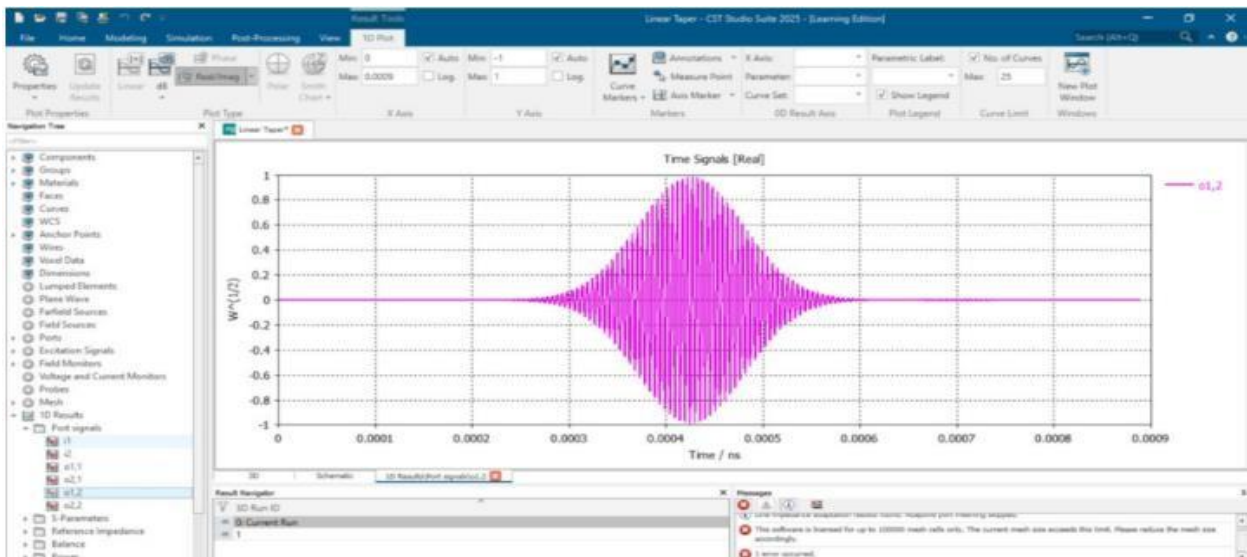
12. Performance Evaluation of a Linear Tapered Rectangular Waveguide in terms of power distribution analysis per ports and S-parameters.

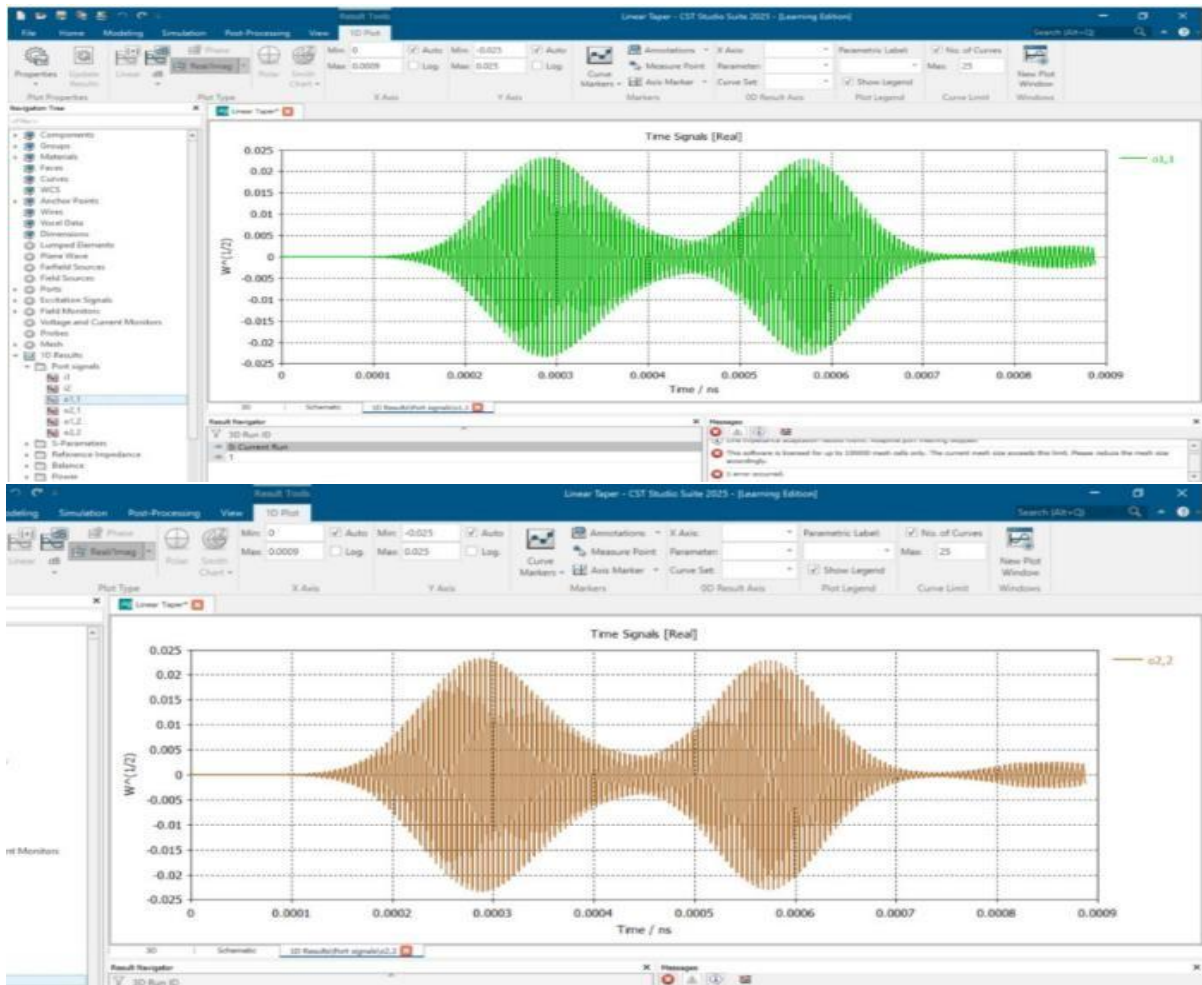
Test Case 1:

Simulate a linear tapered Rectangular Waveguide using CST Studio Suite, analyze its electromagnetic behavior, and examine the power distribution among ports over a specified frequency range.



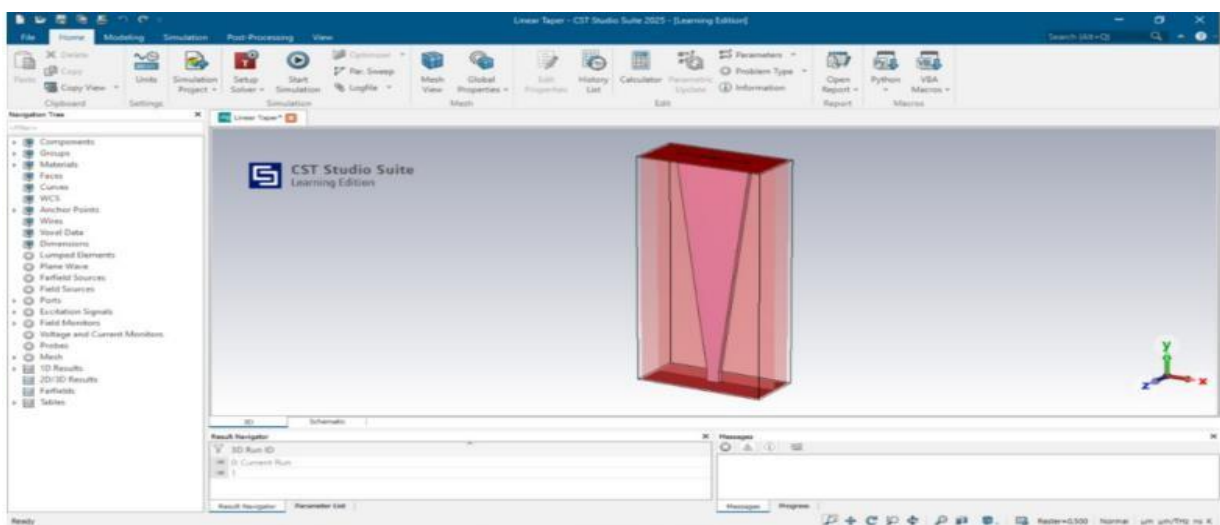
Output:



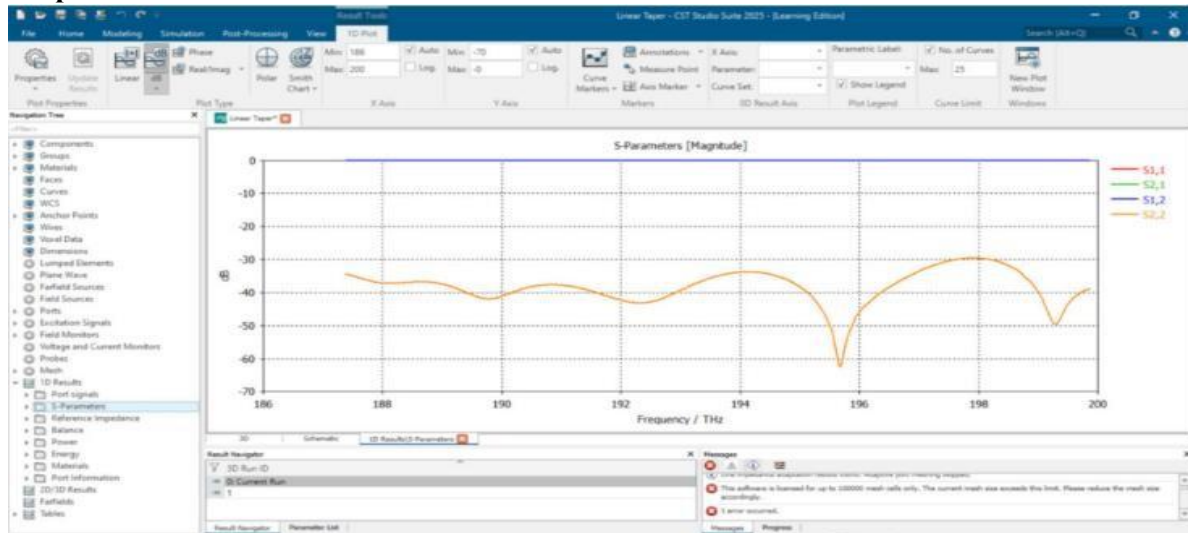


Test Case 2:

Simulate a linear tapered Rectangular Waveguide using CST Studio Suite, analyze its electromagnetic behavior, and examine the S parameters.



Output:



RESULT:

The coaxial connector was successfully simulated using CST Studio Suite over the specified frequency range. The electromagnetic behavior of the structure was analyzed through power distribution among ports shows that the majority of the input power is transmitted to the output port with minimal reflection and negligible losses, indicating efficient signal transmission and satisfactory connector performance.